

Made by-Parth

EduMind

A learning system that remembers how you learn.

An AI Teaching Operating System

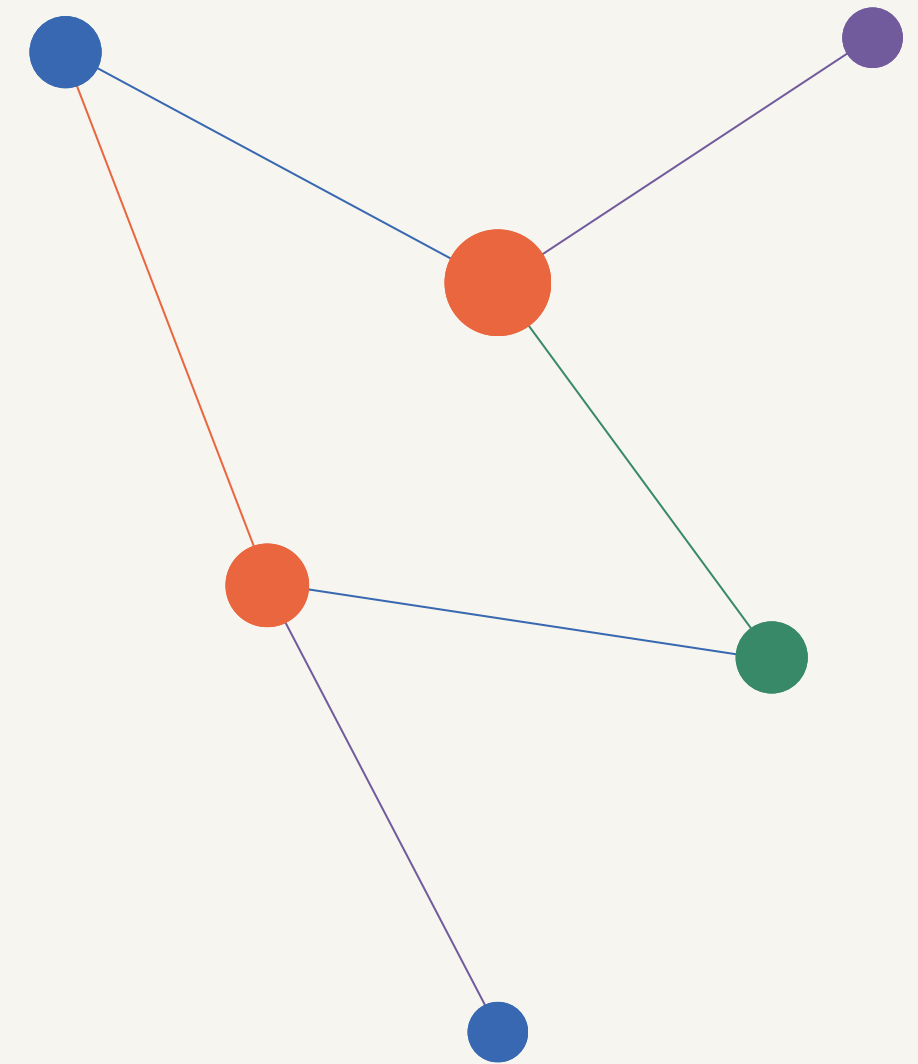
Instead of another chat window, EduMind keeps a persistent model of the learner, adapts the lesson, and uses the result to shape the next session.

Next.js

FastAPI

LangGraph

Postgres



The problem isn't lack of AI.

It is lack of continuity.

A student jumps between tools

Syllabus	What should I study?
YouTube / ChatGPT	How do I learn it?
Practice paper	Did I understand it?
New chat	Why am I stuck?
Google Images	Can I see it?
Notes again	What should I revise?

Every tool solves one moment.

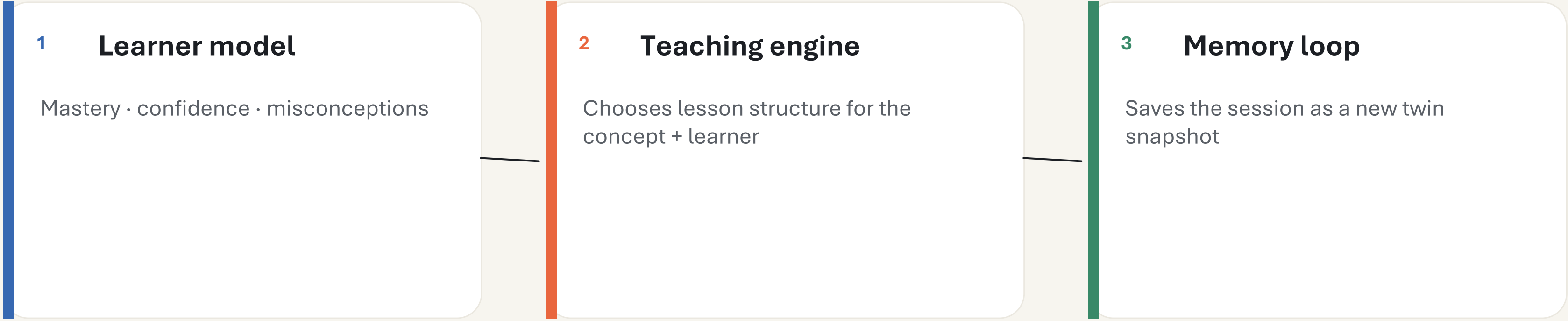
But none keeps a reliable picture of:

-  **What I know** mastery changes over time
-  **What I get wrong** the misconception behind a mistake
-  **What I need next** prerequisites + retention

The result: the student keeps rebuilding context from scratch.

Treat learning like a system, not a chat.

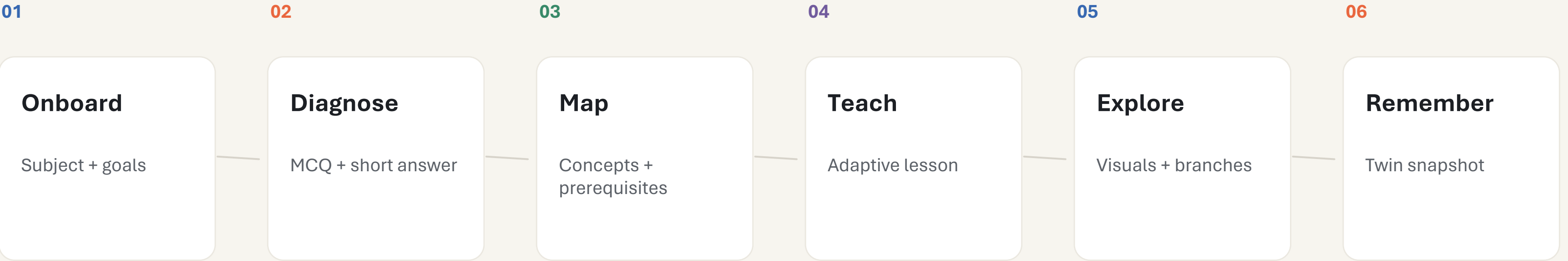
Three pieces work together on every session.



Next session starts with what the system learned last time.

One session, end to end

The student experiences a simple flow; the system handles the complexity.



The important part is the return: the twin changes how the next lesson is taught.

Four features that turn the idea into a system.

01 Digital twin

Versioned learner state: mastery, confidence and misconceptions.

02 Adaptive teaching

Lesson structure is designed for the concept and the learner.

03 Real branching

A side question creates a real checkpoint fork — explore without losing the lesson.

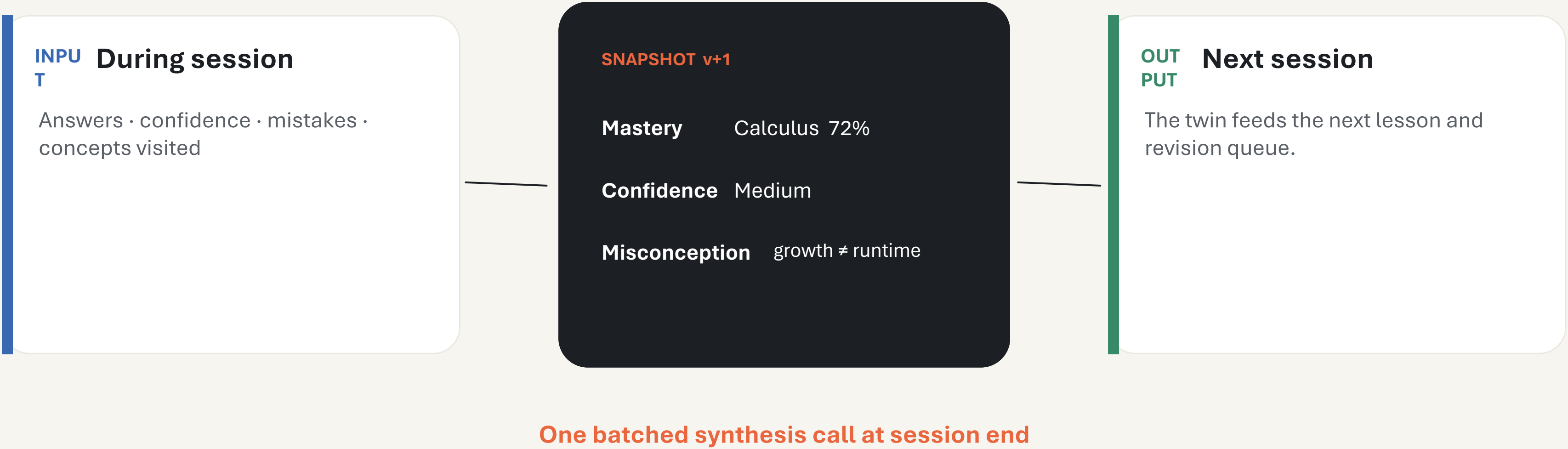
04 Visual layer

An agent decides whether an image, simulation or neither is actually useful.

These are architectural decisions, not just prompt instructions.

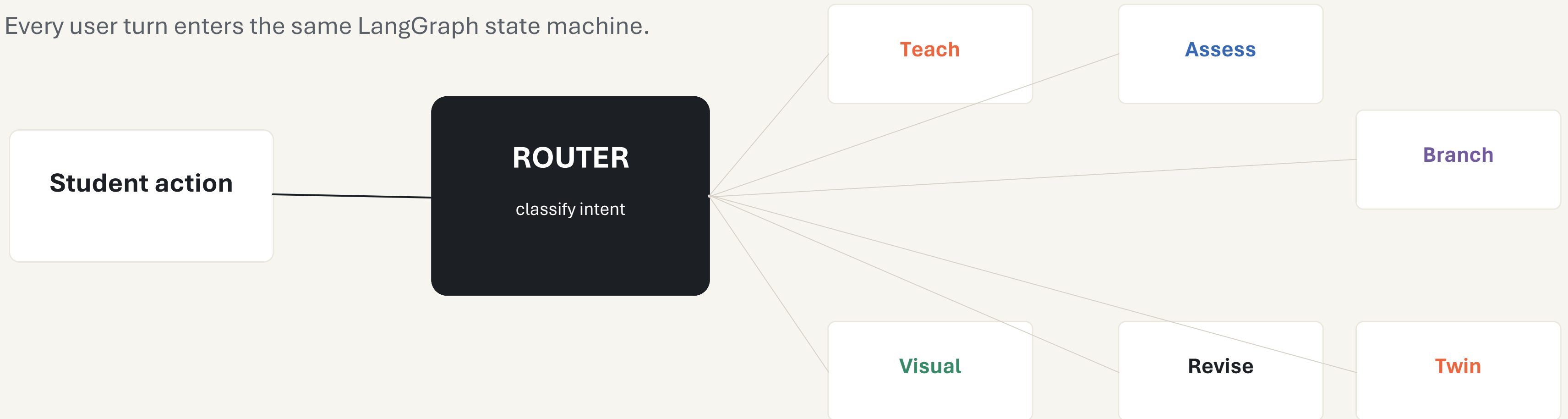
The chat log is not the learner model.

EduMind turns activity into a versioned profile.



One router decides what happens next.

Every user turn enters the same LangGraph state machine.



Some paths do not use an LLM at all.

Revision scheduling + MCQ grading + planning are deterministic.

Cost was a design constraint, not an afterthought.

Continuous tutoring only works if most turns are cheap.

- 01 Cheap on the hot path** Intent classification uses the cheapest capable model.
- 02 Deterministic where possible** MCQ grading, revision and planning need no model.
- 03 Cache before compute** Vision, lessons and simulations are reused.
- 04 Batch expensive work** Twin synthesis runs once at session end.

Model routing is centralized in one function — feature code declares a task, not a model.

Four layers. Four jobs.

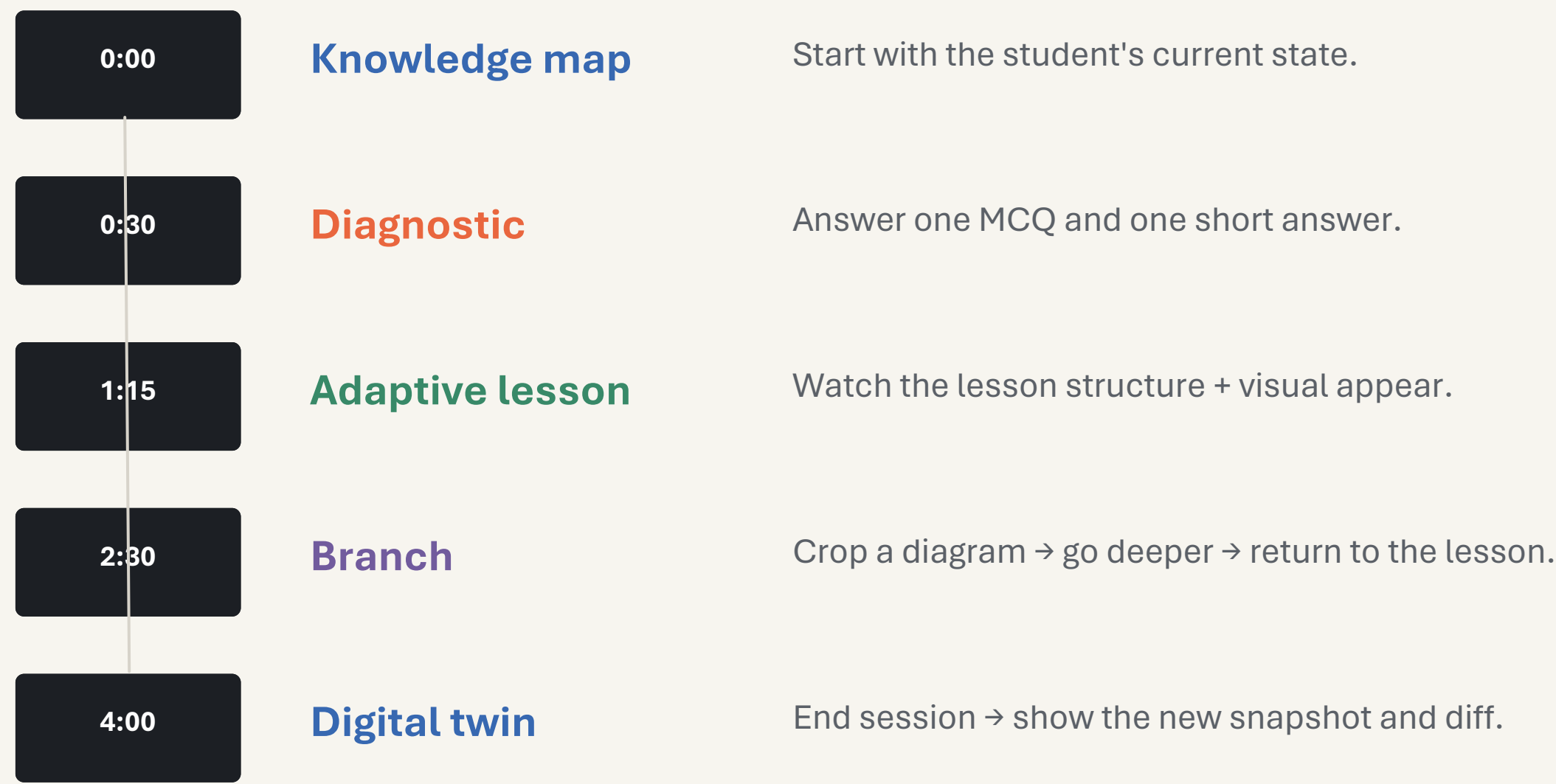
Simple enough to explain; strong enough to run the system.

PRESENTATION	Next.js 16 · React 19 · TypeScript	Learn workspace · knowledge map · twin dashboard
ORCHESTRATION	FastAPI · LangGraph · SSE	StateGraph · router · agents · checkpointed threads
INTELLIGENCE	GPT-5-mini · Claude-Sonnet-4.5 · Gemini-2.5-Flash	Task-based model routing
PERSISTENCE	Postgres 16 · pgvector · SQLAlchemy	Learner state · sessions · documents · snapshots · caches

State lives in the database, not in a context window.

A five-minute story is better than a feature list.

Show one learner moving through the system.



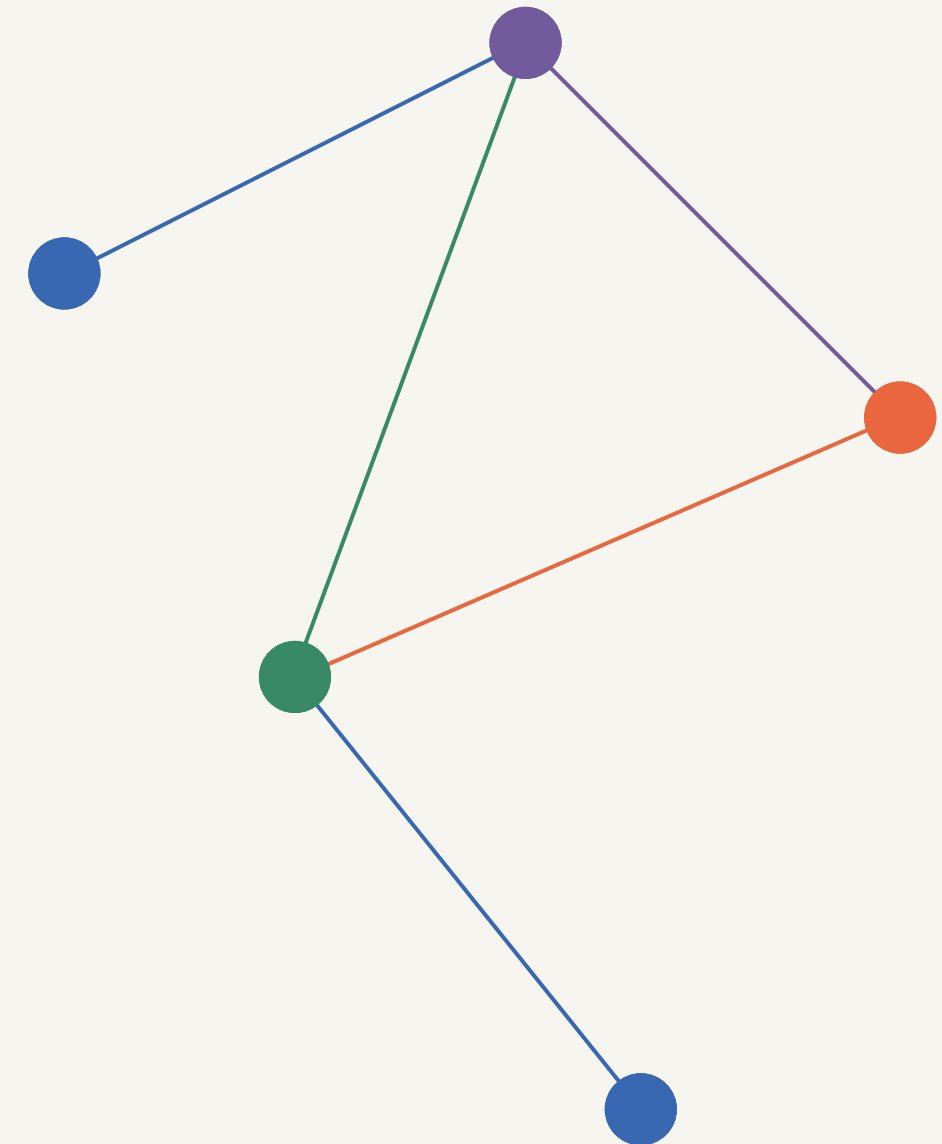
Tip: demo the working product; don't narrate every backend detail.

EduMind

Every AI tutor can answer a question.

EduMind remembers the student.

That memory changes what happens next.



MADE BY

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Thank you • Questions?